# Velocity and position graphs

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Connect a position-time graph to velocity using slope.

**SUCCESS CRITERIA - I can...**

1. Read positions and elapsed times from a graph.
2. Find velocity from position change divided by time.
3. Explain the difference between speed and velocity.

**Core:** teacher selects from 1-9. **Stretch:** 10 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Bring back an earlier idea

East is positive. Start at 5 m, go to -3 m, then finish at 2 m. Sketch the route; calculate distance and displacement. Distinguish both from final position.

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** $v_{avg} = \Delta x/\Delta t$; average speed = distance/$\Delta t$. $\Delta$ means change; $\Delta t$ is elapsed time. Position-time slope gives velocity.

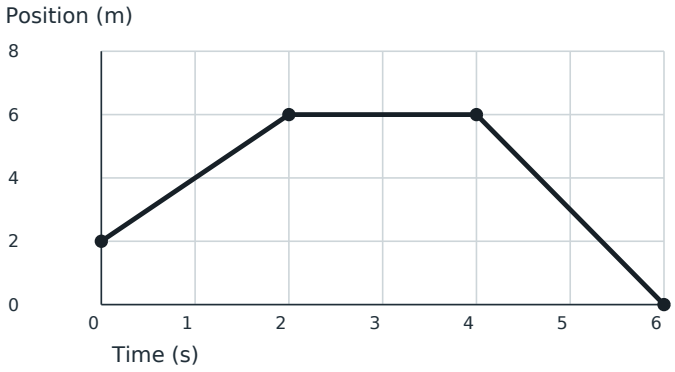
**Worked reasoning:** A cart moves from 1 m to 7 m in 3 s at steady velocity. The position-time slope is (7-1)/3=2 m/s.

**Your turn:** Sketch the graph through the two points. Draw and label the rise and run used for its slope.

**Marzano check - circle:** ① ② ③

## 03 Interpret all three intervals

For the displayed position-time graph, calculate velocity on each segment. Label the direction of motion on each.




**Marzano check - circle:** ① ② ③

## 04 Compare two averages

For the full 6 s trip in 3, calculate average speed and average velocity. Explain their difference using the route.

**Marzano check - circle:** ① ② ③

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Continue down each column.

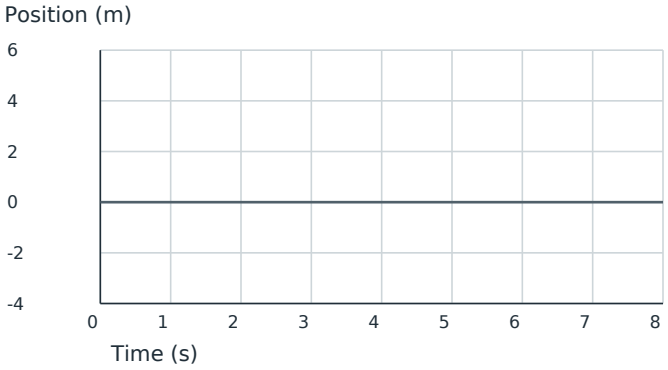
### 05 A line that misses the origin

A straight graph passes through (2 s, 14 m) and (6 s, 6 m). Calculate slope, write x(t), and predict x at 9 s.

**Marzano check - circle:** ① ② ③

### 06 Construct the graph

Start at -2 m. Move at +3 m/s for 2 s, rest for 2 s, then move at -2 m/s for 3 s. Draw a labeled position-time graph; use additional graph paper if needed.




**Marzano check - circle:** ① ② ③

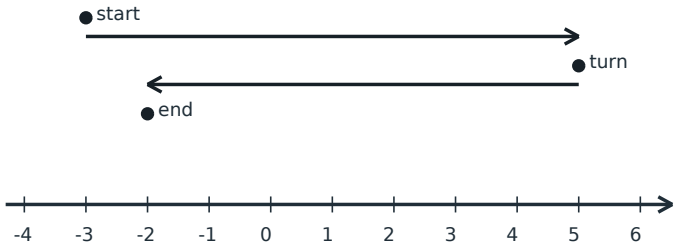
### 07 Separate height and slope

A student reads a point at (5 s, 10 m) and reports a velocity of 2 m/s. Under what additional condition would that calculation be justified?

**Marzano check - circle:** ① ② ③

### 08 Choose a method

East is positive. Start at -3 m, go to 5 m, then finish at -2 m. Sketch the route; calculate distance and displacement. Distinguish both from final position.




Position (m); right is positive

**Marzano check - circle:** ① ② ③

### 09 Exit check

A straight segment runs from (1 s, -2 m) to (5 s, +6 m). Find velocity and state what the slope units mean. Predict x at 7 s if the same motion continues.

**Marzano check - circle:** ① ② ③

### 10 STRETCH: Apply it another way

Two carts follow $x_A = 1 + 2\,t$ and $x_B = 9 - t$, in meters and seconds. Sketch both graphs and find when and where they meet.

**Marzano check - circle:** ① ② ③